Smart Pantry

High Level Design Document

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Introduction

Project Concept / Pitch

Smart Pantry is a pantry management and AI-assisted meal recommendation system designed to help study participants understand what food they already have, use ingredients before they expire, and make practical meal decisions based on their actual pantry inventory. The application is not just a recipe viewer. It is a research-focused software system that connects pantry awareness, recommendation usefulness, and ingredient utilization into one testable platform.

Updated Pitch: Smart Pantry helps a pantry think ahead by tracking usable ingredient amounts, warning users before food expires, and recommending meals that can realistically be made with the food available.

Milestone 3 Scope: This milestone focuses on participant login, participant-specific pantry data, editable quantity tracking, expiration alerts, AI-assisted meal recommendations, ingredient usage logs, pre-study and post-study surveys, and admin exports. Grocery list upload, barcode scanning, and external food database integrations are planned as future enhancements rather than required Milestone 3 features.

Technical Design Goals

The main technical goal of Smart Pantry is to create an intelligent, research-ready pantry management system that supports realistic ingredient tracking and explainable meal recommendations. Earlier versions focused mainly on pantry entry and recipe matching. The improved Milestone 3 design expands the system so pantry items are stored by usable quantity instead of only by item name. For example, a carton of eggs can be stored as individual eggs, bread can be stored as slices, and milk can be stored by cups or another selected unit. When a participant uses a meal recommendation, the system subtracts only the amount used instead of removing the full pantry item.

The system also supports a participant study workflow. Users can register, log in, complete surveys, enter profile preferences, manage their own pantry, view recommendations, mark meals or ingredients as used, and create usage data for analysis. The administrator/researcher view supports study oversight by displaying participant records, pantry activity, recommendation logs, ingredient usage logs, survey responses, and CSV export options.

The recommendation logic uses an explainable weighted scoring approach. It considers ingredient availability, quantity sufficiency, substitution options, expiration urgency, allergies, disliked ingredients, and preferred meal types. This keeps the design understandable for users and defendable for a capstone project while still demonstrating intelligent software behavior.

Development APIs / Programming Languages

Technology	Current Use in Smart Pantry
Python 3.11	Primary programming language for application logic, database operations, recommendation scoring, and data handling.
Streamlit	Web interface for login, pantry pages, surveys, recommendation display, admin dashboard, and interactive forms.
SQLite	Local relational database used for participant accounts, pantry items, recommendation logs, ingredient usage logs, and surveys.
Pandas	Data processing for tables, admin views, survey exports, and future CSV upload workflows.
scikit-learn (future)	Potential future ingredient vectorization, similarity scoring, clustering, or personalization experiments.
USDA FoodData Central / Open Food Facts (future)	Possible future ingredient metadata, nutrition, barcode, and serving-size support. These are outside required Milestone 3 scope.

Development Standards / Cold-Start Procedures

The project is managed through GitHub and Trello. The Trello board should reflect the current Smart Pantry direction and should use lists such as Project Backlog, Milestone 3 - This Week, In Progress, Testing / Review, Completed, Bugs / Risks, and Future Work.

Development is completed in Visual Studio Code using Python 3.11. The project should be opened from the main Smart Pantry folder. Developers should create and activate a virtual environment, install dependencies from requirements.txt, and run the Streamlit application from the project root.

Cold-Start Step	Command or Action
Open project	Open the Smart Pantry folder in Visual Studio Code.
Create environment	python -m venv .venv
Activate environment	Windows: .venv\Scripts\activate

Install dependencies	pip install -r requirements.txt
Run application	streamlit run main.py
Database startup	SQLite tables are initialized automatically when the app starts.

If an older local smart_pantry.db file causes missing-column errors during testing, the database can be regenerated after confirming that no needed test data will be lost. This is a development-only reset step, not a participant data management procedure.

System Architecture

Mode	Main Access	Purpose

Participant Mode	Regular participant login	Allows participants to manage their own pantry, complete surveys, view

Smart Pantry uses a layered architecture so the user interface, application logic, data processing, database storage, and research tools can be understood separately. This improves maintainability and makes the design easier to explain during milestone review.

Layer	Component	Purpose
Presentation Layer	Streamlit UI	Displays login, home dashboard, profile, surveys, pantry management, recommendations, history, and admin pages.
Application Logic Layer	Python functions	Handles authentication workflow, validation, recommendation scoring, expiration alerts, quantity deduction, and study workflows.
Recommendation Layer	Explainable scoring engine	Ranks meals based on pantry availability, quantity sufficiency, expiration urgency, substitutions, and user preferences.
Data Processing Layer	Pandas	Supports table handling, export preparation, survey analysis preparation, and future upload processing.
Data Storage Layer	SQLite	Stores users, pantry items, recommendation logs, ingredient usage logs, and survey responses.
Research/Admin Layer	Admin dashboard and CSV exports	Allows the researcher to review participant data and export records for final capstone analysis.

Project Functionality

User Modes

		recommendations, and mark meals or ingredients as used.
Administrator / Researcher Mode	Hidden admin access	Allows the researcher to review participant activity, view survey results, check logs, and export data. Admin credentials should not be displayed publicly on the login page.
Participant Data Separation	user_id-based records	Ensures participants only see their own pantry, survey responses, recommendation history, and usage logs while the admin can review the full study dataset.

Main Mechanics / Sub Mechanics

Mechanic	Description	Milestone 3 Decision
Usable Quantity Tracking	Each pantry item stores quantity, unit, container type, category, expiration date, and user_id.	Required for Milestone 3.
Editable Serving Defaults	Common items can start with household defaults that the participant can edit before saving.	Required because it reduces user burden while keeping inventory realistic.
Expiration Alerts	Dashboard warns users based on days left before expiration.	Required; red, orange, and baby blue alerts support quick decisions.
Recommendation Scoring	The system ranks meals using pantry match, quantity sufficiency, expiration urgency, substitutions, and preferences.	Required; this is the core intelligent behavior.

Substitution Suggestions	When a required ingredient is missing, the app can suggest a reasonable substitute if available.	Required as a practical recommendation feature.
Allergy and Preference Warnings	The app warns users when a recipe contains an allergy or disliked ingredient.	Required; warnings should guide the user without automatically removing every possible meal.
Partial Ingredient Deduction	When a meal is marked as used, only the recipe-required amount is subtracted from the pantry.	Required; prevents unrealistic deletion of entire pantry items.
Study Data Collection	Recommendation actions, usage events, surveys, and pantry activity are stored for analysis.	Required for capstone research and final reporting.
Grocery List Upload	CSV/TXT upload for adding multiple items at once.	Moved to Milestone 4 / Future Work to keep Milestone 3 realistic.

AI Module(s)

The AI module is best described as an AI-assisted recommendation engine. It does not need to be a black-box neural network to be intelligent. The intelligence comes from combining multiple contextual factors and producing a ranked recommendation that is more useful than a generic recipe search.

Scoring Factor	Purpose	Example System Behavior
Ingredient availability	Checks whether required recipe ingredients exist in the participant pantry.	A meal is only shown when the user has the needed ingredients or approved substitutes.
Quantity sufficiency	Confirms the pantry has enough usable amount for the recipe.	A recipe needing 2 eggs requires at least 2 eggs available.
Expiration urgency	Prioritizes meals that use ingredients close to expiration.	Food with 0-4 days left can increase recommendation priority.
Substitution availability	Allows practical alternatives when safe and reasonable.	Shrimp may be suggested as a substitute for chicken if the recipe still works.
Allergy / dislike warnings	Flags ingredients that may need substitution or removal.	A recipe containing onion is flagged for the user instead of silently ignored.
Meal type preference	Ranks meals closer to the participant	Breakfast, lunch, dinner, or brunch
	selected preferences.	preferences influence ranking.

The recommendation score should be explainable in the interface. Each recommendation should tell the user why it was suggested, such as matched ingredients, expiring ingredients, substitutions, and warnings. This makes the system testable during the study and easier to defend during project review.

User Interface / Inputs / Sensors

The interface is built in Streamlit and organized into study-focused pages. Participant navigation should include Home, Profile, Pre-Study Survey, My Pantry, Meal Recommendations, Recommendation History, Post-Study Survey, and Logout. Admin navigation should include Admin Dashboard, Participant View, CSV Exports, and Logout.

The homepage should use Smart Pantry branding and the project logo when available. The previous extra marketing sentence should remain removed. The participant login page should not reveal admin username or password information. The visual direction should use green for pantry/health identity, orange for urgency/action, and blue or baby blue for lower-priority alerts and interface accents.

Inputs include text fields, dropdowns, date selectors, editable tables, buttons, sliders, and survey forms. Milestone 3 should prioritize manual pantry entry and editable serving defaults. Grocery list upload can remain documented as a Milestone 4 enhancement.

Graphics / Animation / Video

Smart Pantry does not require advanced animation or video for Milestone 3. The visual design focuses on readability, dashboard cards, tables, color-coded alerts, and food-related branding. Future versions may include food thumbnails, richer charts, and a more mobile-friendly layout.

Audio

The current version does not require audio. Expiration warnings and pantry updates are shown visually on the dashboard. Optional notification sounds are outside the Milestone 3 scope.

Networking

The current application operates as a local Streamlit web app with SQLite data storage. It can be run without a paid website or domain, which supports the current user study scope. Future versions can move to cloud hosting, PostgreSQL, FastAPI services, barcode lookup, Open Food Facts, USDA FoodData Central, or other external services after the Milestone 3 workflow is stable.

Databases / Server-Side Technology

Smart Pantry requires a database because it stores participant accounts, pantry items, recommendation logs, ingredient usage logs, and survey responses. SQLite is used for Milestone 3 because it is lightweight, local, and appropriate for prototype testing.

Table	Purpose	Important Fields
users	Stores participant and admin accounts plus profile preferences.	id, username, password_hash, role, allergies, disliked_ingredients, preferred_meal_types
pantry_items	Stores each participant pantry item.	user_id, item_name, category, quantity, unit, container_type, expiration_date, status
meal_recommendation_logs	Stores viewed, saved, and used recommendations.	user_id, meal_name, meal_type, score, matched_ingredients, expiring_ingredients, used_recommendation, feedback, created_at
ingredient_usage_logs	Stores ingredient usage events and remaining amounts.	user_id, pantry_item_id, item_name, usage_type, quantity_used, unit, remaining_quantity, notes, created_at
surveys	Stores pre-study and post-study survey responses.	user_id, survey_type, pantry_awareness, recommendation_usefulness, ingredient_utilization, ease_of_use, current_method, comments, created_at

The quantity-based design is a major correction. Instead of marking an item as fully used after one meal, the database records how much was used and how much remains. This supports realistic study data because partial usage, such as using part of a carton or part of a loaf, can be measured over time.

Localization Plan

The MVP is focused on English-language users in the United States. Localization is not a Milestone 3 requirement. Future localization may include multilingual interface text, regional measurement conversions, and culture-specific ingredient names.

Custom Topics

A custom design focus for Milestone 3 is editable household serving defaults. This approach is more realistic than requiring participants to read every package label while still allowing correction before saving. Examples include eggs stored as individual eggs, bread stored as slices, milk stored by cups, cheese stored by slices, and tortillas stored by pieces. Future improvements can connect these defaults to barcode scanning or food databases.

Another custom topic is research data collection. Smart Pantry is built for participant use and for capstone analysis. The admin dashboard and exports support later analysis of pantry awareness, recommendation usefulness, and ingredient utilization.

Research Data Collected

Data Type	Why It Matters
Pantry items entered	Measures whether participants are becoming more aware of what food they already have.
Expiration alerts displayed	Shows whether users are being exposed to reminders that may encourage food use before expiration.
Recommendations generated	Measures whether the system is producing practical meal suggestions from pantry data.
Recommendations saved or used	Shows which recommendations participants found useful enough to act on.
Ingredient usage deductions	Tracks partial ingredient use more realistically than deleting full items.
Allergy and substitution warnings	Shows how often recommendations need safety or preference guidance.
Pre-study and post-study surveys	Measures changes in pantry awareness, recommendation usefulness, ingredient utilization, and ease of use.
Admin CSV exports	Supports final analysis and reporting without requiring the participant to see research-only dashboards.

Design Consistency and Interface Flow

To keep the design consistent with the High Level Design Document format, Smart Pantry follows the same type of structured flow used in the provided design documentation template: concept, technical goals, functionality, interface flow, production documentation, roadmap, testing, and references. The added consistency focus explains how the user experience, visual style, data rules, and research workflow should remain stable across the application.

Consistency Area	Design Rule	Purpose
Navigation Consistency	Participant pages should follow the same order: Home, Profile, Pre-Study Survey, My Pantry, Meal Recommendations, Recommendation History, Post-Study Survey, and Logout.	Keeps the participant study flow easy to follow from login through final survey.
Visual Consistency	Use green for pantry/health identity, orange for Warning: Expires Soon, baby blue for Use Soon, and red for Expired / Spoiled Check.	Prevents confusing alert meanings and supports quick decision-making.
Interaction Consistency	Buttons, tables, forms, and confirmation messages should behave the same way across pantry entry, recommendations, surveys, and admin exports.	Reduces participant confusion and makes testing more reliable.
Recommendation Consistency	Each recommendation should show why it appeared, including matched ingredients, quantity status, expiration urgency, substitutions, and warnings.	Keeps the AI-assisted logic explainable and defendable.
Data Consistency	All pantry items, surveys, recommendations, and usage logs should stay tied to user_id records.	Protects participant data separation and supports accurate research exports.
Documentation Consistency	Feature names, alert labels, table names, and milestone scope should stay the same in the app, Trello board, and design document.	Makes the project easier to present, test, and defend during milestone review.

This consistency layer strengthens the design by showing that Smart Pantry is not only a collection of features. It is a structured participant workflow where the interface, alert logic, recommendation behavior, database records, and research outputs all support the same purpose.

Mockups / Flowcharts

The updated Smart Pantry interface follows a multi-page Streamlit navigation model. Participant users see the pages needed for study participation, while the administrator sees dashboard and export tools. The app homepage includes branding, summary metrics, and expiration alerts. The pantry page includes manual item entry, editable serving defaults, current pantry items, partial ingredient use, and delete options. The recommendation page displays meals that match inventory and quantity requirements.

Participant Navigation	**Admin Navigation**
Home	Admin Dashboard
Profile	Participant View
Pre-Study Survey	CSV Exports
My Pantry	Logout
Meal Recommendations	
Recommendation History	
Post-Study Survey	
Logout	

Application Flow

1. Participant creates an account or logs in.
2. Participant completes profile preferences, including allergies, disliked ingredients, and preferred meal types.
3. Participant completes the pre-study survey.
4. Participant adds pantry items using manual entry or editable household defaults.
5. System stores quantity, unit, container type, expiration date, and user_id in SQLite.
6. Dashboard displays pantry metrics and expiration alerts using Expired / Spoiled Check, Warning: Expires Soon, and Use Soon cards.
7. Recommendation engine checks ingredient availability, quantity sufficiency, substitutions, preferences, and expiration urgency.
8. Participant saves a recommendation or marks a meal as used.
9. When a meal is used, the system subtracts only the recipe-required ingredient amounts.
10. Ingredient usage logs and recommendation logs are stored for admin review.
11. Participant completes the post-study survey.
12. Admin exports users, pantry data, recommendation logs, ingredient usage logs, and survey responses for analysis.

Alert Status	Days Left	Color	Purpose
Expired / Spoiled Check	0-1 days	Red	Immediate attention. The item is expired, at expiration, or close enough that the participant should check it before using.
Warning: Expires Soon	2-4 days	Orange	The item expires soon and should be used before it goes bad.

Use Soon	5-10 days	Baby blue	The item is getting close to expiration, so the participant should consider planning a meal with it.
No Alert	11+ days	No alert card	The item is not close enough to expiration to display an alert.

Step	System Behavior
Ingredient Check	The app checks whether each required recipe ingredient exists in the participant pantry.
Quantity Check	The app verifies that enough usable quantity exists for each required ingredient.
Substitution Check	If a required item is missing, the system checks approved substitutes.
Warning Check	The app displays allergy or disliked ingredient warnings from the participant profile.
Recommendation Display	Only complete or substitutable meals are shown to the participant.
Meal Used	The app logs the recommendation as used and subtracts recipe quantities from pantry items.
Item Empty	If a pantry item reaches zero, it is removed from the active pantry list or marked inactive.

Production Documentation

Feature List

Number	Feature	Category	Priority	Scope
F1	Participant Login and Registration	UI/System	High	Milestone 3
F2	Hidden Admin Login Access	Security/System	High	Milestone 3
F3	Participant Profile Preferences	UI/System	High	Milestone 3
F4	Pre-Study Survey	Research/UI	High	Milestone 3
F5	Post-Study Survey	Research/UI	High	Milestone 3
F6	Manual Pantry Item Entry	UI/System	High	Milestone 3
F7	Editable Serving Defaults	System	High	Milestone 3
F8	Quantity, Unit, and Container Tracking	Database/System	High	Milestone 3
F9	Partial Ingredient Usage Deduction	System	High	Milestone 3
F10	Expiration Alert Dashboard	UI/System	High	Milestone 3
F11	Expired / Warning / Use Soon Alert Rules	UI/System	High	Milestone 3
F12	AI-Assisted Meal Recommendation Engine	System	High	Milestone 3
F13	Pantry-to-Recipe Matching	System	High	Milestone 3

F14	Quantity Sufficiency Checks	System	High	Milestone 3
F15	Substitution Suggestions	System	Medium	Milestone 3
F16	Allergy and Preference Warnings	System/UI	High	Milestone 3
F17	Recommendation Scoring and Explanation	System	High	Milestone 3
F18	Recommendation History	Research/System	Medium	Milestone 3
F19	Ingredient Usage Logs	Research/Database	High	Milestone 3
F20	Admin Dashboard	Research/UI	High	Milestone 3
F21	CSV Data Exports	Research/System	High	Milestone 3
F22	Green / Orange / Blue Styling	UI	Medium	Milestone 3
F23	Homepage Logo / Branding Support	UI	Medium	Milestone 3
F24	SQLite Database Integration	Database	High	Milestone 3
F25	User-Specific Pantry Separation	Database/System	High	Milestone 3
F26	Pandas Data Processing	Tool	High	Milestone 3
F27	Streamlit Web Interface	Tool	High	Milestone 3
F28	GitHub Version Control	Tool	High	Milestone 3
F29	Trello Project Tracking	Project Management	High	Milestone 3
F30	Grocery List Upload - CSV/TXT	UI/System	Low	Milestone 4 / Future
F31	Open Food Facts	Future Work	Low	Future
	Integration			
F32	USDA FoodData Central Integration	Future Work	Low	Future
F33	Barcode Scanning	Future Work	Low	Future
F34	Cloud Deployment	Future Work	Low	Future
F35	Design Consistency and Interface Flow Rules	UI/System	Medium	Milestone 3

Master Task List

Number	Task	Feature(s)	Dependencies
T1	Clean Trello board and align cards with Smart Pantry Milestone 3.	F29	NA
T2	Update project title and documentation to Smart Pantry only.	F28-F29	T1
T3	Implement participant login and account workflow.	F1	NA
T4	Keep admin login functional while removing public credential display.	F2	T3

T5	Update profile preferences for allergies, disliked ingredients, and preferred meal types.	F3	T3
T6	Implement pre-study and post-study surveys.	F4-F5	T3
T7	Develop SQLite schema for users, pantry, surveys, recommendations, and usage logs.	F24-F25	NA
T8	Add container_type, quantity tracking, and partial usage fields.	F7-F9	T7
T9	Create manual pantry item entry with quantity, unit, container, category, and expiration date.	F6-F8	T8
T10	Add editable serving defaults for common household items.	F7	T9
T11	Implement expiration alert rules for Expired / Spoiled Check, Warning: Expires Soon, and Use Soon alerts.	F10-F11	T9
T12	Update homepage with logo support and remove unwanted marketing text.	F22-F23	T3
T13	Update recommendation logic to check ingredient existence and quantity sufficiency.	F12-F14	T9
T14	Implement substitutions and profile warning logic.	F15-F16	T13
T15	Add recommendation save and used-meal workflows.	F17-F18	T13
T16	Subtract recipe ingredient amounts when meal is marked used.	F9-F19	T15
T17	Update admin dashboard tables and exports.	F20-F21	T7
T18	Test participant workflow from registration to poststudy survey.	F1-F21	T3-T17
T19	Test admin workflow and CSV exports.	F20-F21	T17
T20	Move grocery list upload documentation to Milestone 4 / Future Work.	F30	T2
T21	Document future API, barcode, cloud, and nutrition enhancements.	F31-F34	T17
T22	Review interface, alert labels, and documentation wording for design consistency	F35	T3-T21

Development Roadmap

Milestone	Planned Work
Milestone 1 - Foundation and Core System Development	Set up the Python/Streamlit project, SQLite database, GitHub repository, Trello board, basic login, basic pantry entry, and initial navigation. Establish the base architecture for storing pantry data and participant information.
Milestone 2 - Recommendation Engine and Pantry Workflow	Build pantry-to-recipe matching, weighted scoring, expiration logic, and recommendation display. Confirm that meals are ranked using pantry inventory instead of generic recipe browsing.
Milestone 3 - Participant Study Readiness	Implement participant login, profile preferences, surveys, user-specific pantry records, editable quantity tracking, partial ingredient deduction, recommendation logs, ingredient usage logs, admin dashboard, and CSV exports.
Milestone 4 - Expansion and Polish	Add grocery list upload, barcode scanning research, external food database connections, stronger analytics visualizations, cloud deployment planning, and mobile-friendly improvements if time allows.

Verification and Validation

Verification confirms that Smart Pantry was built correctly. Validation confirms that the system supports the actual research purpose: improving pantry awareness, recommendation usefulness, and ingredient utilization. The Milestone 3 V&V plan focuses on participant workflow, data separation, recommendation behavior, partial ingredient use, and admin data exports.

V&V Area	Verification Question	Validation Question
Authentication	Can participants and the admin log in through the correct workflow?	Does the login structure support a real participant study?
Data Separation	Does every pantry, survey, and recommendation record store the correct user_id?	Can participants safely use the app without seeing other participant data?
Quantity Tracking	Does the app correctly store and update quantity, unit, and container type?	Does quantity tracking reflect how food is used in real life?
Recommendations	Does the system check availability, quantity, substitutions, warnings, and expiration urgency?	Do the recommendations feel practical enough for participants to use?
Research Logs	Are recommendation and ingredient usage events saved correctly?	Can the researcher analyze usage behavior after the study?
Surveys	Can users complete pre-study and post-study surveys?	Do the surveys capture changes in awareness, usefulness, and behavior?
Admin Exports	Can the admin export usable CSV files?	Can exported data support final capstone analysis?

Quality Test List

Test ID	Test Case	Expected Result
Q1	Verify participant data separation.	A participant can only view their own pantry, survey responses, recommendation history, and usage logs.
Q2	Verify admin research access.	Admin can view participant records, pantry activity, surveys, recommendation logs, usage logs, and export files.
Q3	Verify hidden admin login behavior.	Admin access remains functional, but admin credentials are not publicly displayed on the participant login screen.

Q4	Verify manual pantry entry.	User can add an item with item name, category, quantity, unit, container type, and expiration date.
Q5	Verify editable serving defaults.	Common items load helpful default quantities, and the user can edit the amount before saving.
Q6	Verify partial ingredient deduction.	When a meal uses part of an ingredient, the remaining pantry quantity updates correctly.
Q7	Verify zero-quantity handling.	When an ingredient reaches zero, it is removed from the active pantry list or marked inactive.
Q8	Verify expiration alert logic.	Items display the correct Expired / Spoiled Check, Warning: Expires Soon, Use Soon, or No Alert status based on days left.
Q9	Verify expiration-based recommendation priority.	Meals using soon-to-expire ingredients receive higher recommendation priority when other requirements are met.
Q10	Verify recommendation completeness.	Meals only appear when required ingredients or approved substitutes are available in enough quantity.
Q11	Verify substitution handling.	When a recipe ingredient is missing but a reasonable substitute exists, the system clearly shows the substitute option.
Q12	Verify allergy warning behavior.	Recipes with allergy ingredients are flagged with a warning and substitution/removal guidance.
Q13	Verify disliked ingredient warning behavior.	Recipes with disliked ingredients are flagged so the user can decide whether to substitute or skip the meal.
Q14	Verify recommendation explanation.	Each recommendation shows why it was suggested, including matched items, expiring items, substitutions, or warnings.
Q15	Verify recommendation logging.	Viewed, saved, or used recommendations are stored with user_id and timestamp.
Q16	Verify ingredient usage logging.	Ingredient usage events store item name, quantity used, unit, remaining quantity, user_id, and timestamp.
Q17	Verify survey workflow.	Participant can complete pre-study and post-study surveys once per account.
Q18	Verify CSV export accuracy.	Admin exports contain correct
		participant activity without mixing user records.
Q19	Verify Streamlit navigation.	Participant and admin pages route correctly based on login role.
Q20	Verify no Milestone 4 feature dependency.	Milestone 3 workflow still works without grocery list upload, barcode scanning, or external APIs.

Future Work

- Grocery list upload through CSV/TXT with review before saving items to the pantry.
- Barcode scanning or receipt scanning to reduce manual entry time.
- Open Food Facts or USDA FoodData Central integration for richer nutrition and serving-size information.

- Cloud deployment with PostgreSQL if the study grows beyond local testing.
- Personalized recommendation learning after enough participant usage data is collected.
- Mobile-friendly interface improvements and optional notification features.

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